PASSWORD-BASED AUTHENTICATION SYSTEM

LPC2129 ARM7 • Embedded C • Proteus Simulation

Project requirements and implementation checklist

This document lists the components, software, suggested pin assignments, and planned behavior for an educational prototype. The first version uses UART input through the Proteus Virtual Terminal, a 16×2 LCD, and two status LEDs. A 4×4 keypad can be added later.

1. Hardware requirements

Component	Purpose	First version?
LPC2129 ARM7 microcontroller	Validates the entered password and controls outputs.	Yes
16×2 character LCD	Displays prompts, access results, and lockout messages.	Yes
UART0 / Proteus Virtual Terminal	Receives password characters and displays status messages.	Yes
Green LED + current-limiting resistor	Indicates access granted.	Yes
Red LED + current-limiting resistor	Indicates access denied.	Yes
4×4 matrix keypad	Alternative way to enter a password using physical/simulated keys.	Optional upgrade
Power and ground	Provides correct supply and reference connections.	Yes

2. Software requirements

Software / file	Purpose
Keil µVision 5	Write, compile, and debug the Embedded C program.
Proteus ISIS	Simulate the microcontroller, LCD, LEDs, and UART terminal.
LPC2129 device support	Configure target device and startup files.
lpc21xx.h	Provides LPC21xx register definitions in the project environment.
Your header.h and existing drivers	Reuse existing LCD, UART, and delay functions.

3. Example connections

These are suggested assignments. Confirm the pins are free and match your existing circuit before wiring.

Signal	Suggested pin / setting	Notes
UART0 TXD0	P0.0	MCU TXD0 → Virtual Terminal RXD.
UART0 RXD0	P0.1	Virtual Terminal TXD → MCU RXD0.
Green LED	P0.8	Example only; add a current-limiting resistor.
Red LED	P0.9	Example only; add a current-limiting resistor.
16×2 LCD	Reuse existing wiring	Current driver uses P1.17–P1.23.

4. Planned functionality

1. Display a prompt asking for the password.

2. Receive entered characters through UART0.

3. Compare the entered password with the configured test password (example: 1234).

4. Correct password: display ACCESS GRANTED and turn the green LED ON.

5. Incorrect password: display ACCESS DENIED and turn the red LED ON.

6. Count consecutive failed attempts; after three failures, apply a temporary lockout.

7. After lockout, allow another attempt and reset the failure count after successful authentication.

5. Suggested test plan

Test	Input	Expected result
Correct password	1234 + Enter	Access granted; green LED ON.
Wrong password	Example: 9876 + Enter	Access denied; red LED ON.
Three failures	Three incorrect entries	Temporary lockout message; reject input during lockout.
Recovery	Wait until lockout expires	Allow a new password attempt.

6. Important implementation notes

• UART input is the simplest first version because it reuses your existing UART0 driver. The keypad is an optional later upgrade.

• Match UART baud rate and peripheral clock to your Proteus configuration.

• Confirm LED polarity and wiring; the suggested pin numbers are examples only.

• This is an educational prototype. A hard-coded password and software-only lockout are not suitable for securing a real door or safety-critical system.

Project scope: UART-based password entry first; 4×4 keypad as an optional extension.